Project Name: SMT for energy usage optimization
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Objective: Constraint satisfaction scheduling for a server, with possible power minimization as secondary objective
We want to schedule jobs so that we minimize power in a server.
We will work with a 24h window and real job dataset from a google server (Borg). We can increase the time window to up to 1 month as needed.
We have 5 constraints:

- Power cannot be more than 80% of server power.
- CPU utilization cannot be above 100% of CPU available on the server.
- A job must run to completion (job must finish before the window is over - start + length < 24h)
- Jobs are categorized into 4 scheduling categories - ranging from 0 (most flexible) to 3 (least flexible). Category 0 start time can be delayed 10 hours, category 1 can be delayed 5 hours, category 2 can be delayed 3 hours, and category 3 can be delayed 1 hour.

The order of the jobs and their concurrency determines the power consumption; power depends on cpu and it follows a non-linear curve described in the Barroso Paper (https://dl.acm.org/doi/10.1145/1250662.1250665). We modify that curve into a 4-step piecewise linear function to maintain linearity for the problem. We know ahead of time what the cpu utilization curve looks like for each job, and we have the cpu-power curve coefficients to correlate server cpu utilization (sum of all jobs cpu at time t) to server power.
We plan to use the z3 solver.
Our objective is to verify that there's a way to order the jobs (i.e. a solution) with a given dataset such that we can satisfy the constraints. If a solution is found, and if time permits, we can also identify which solution minimizes power for the server (the optimization is modelled as a linear program – to check that a solution is minimal, we can check that the values of the dual and primal match). For now, we have five constraints, but we can add more constraints pending time and progress.

Week 5 (10/26)	Meet with Professor, discuss and finalize plan for project
Week 6 (11/02)	Data Collection
Week 7 (11/09)	Mathematical formulation with constraints; project check in with Professor Fremont
Week 8 (11/16)	Build the SMT formula using Z3 and use it to find a solution
Week 9 (11/23)	Verify the solution provided by SMT is correct with our data Review work and update based on feedback Project check-in with Professor Fremont
Week 10 (11/30)	Prepare presentation
Week 11	Write and submit the report. Present on Wednesday